

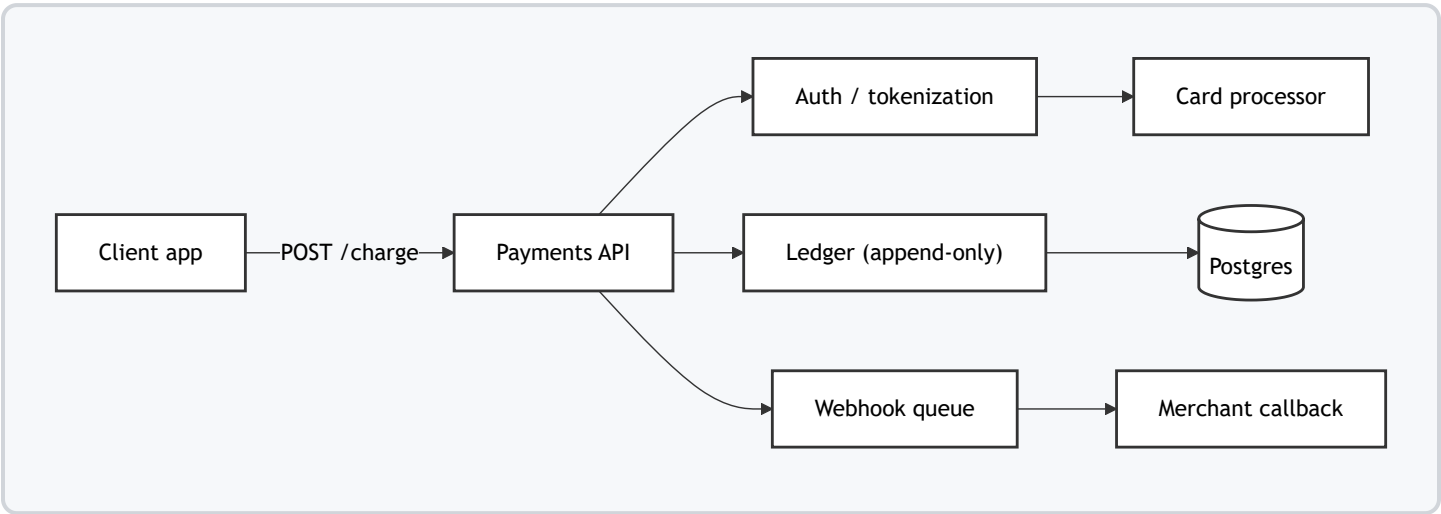
Payments Service — Design Spec

A small, auditable service that authorises card payments and records every state transition.

Note

This document is illustrative. Amounts and identifiers are examples, not production values.

Architecture



Request flow

1. The client posts a charge with an idempotency key.
2. The API tokenizes the card and calls the processor.
3. Every transition is appended to the **ledger** — nothing is ever mutated in place.
4. A webhook notifies the merchant asynchronously.

```
{
  "idempotency_key": "chg_9f2a...",
  "amount": 4200,
  "currency": "AUD",
  "capture": true
}
```

State machine

State	Next	Trigger
created	authorized	processor approves
authorized	captured	capture requested
authorized	voided	timeout / cancel
captured	refunded	refund issued

Warning

Idempotency keys expire after 24h. Re-using an expired key starts a **new** charge.